

CSC3003S Capstone Project — Stage One

Goals (Scope) [21 Marks]

Project Abbreviation and Name	BIOFILM2 Modelling bacterial colonies
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Overall purpose and stakeholders [5]	The aim of this project is to develop a particle-based biophysical model that simulates early-stage biofilm formation, with focus centered on the role of extracellular polymeric substances (EPS) in driving cell aggregation. The model will represent a growing and expanding bacterial colony of mobile, rod-shaped bacteria which can move, reproduce, and secrete EPS. The primary goal is to explore how secreted EPS impacts the clustering of bacterial cells into dense, aggregated structures, which are an important precursor to mature biofilms. The simulation will be implemented in Java as a multithreaded application. It will include a simulation module as well as a separate visualization module for presentation of the results. The model will serve as a research tool for understanding the underlying biophysical mechanisms of biofilm formation, as well as a framework that can potentially be extended to incorporate more complex biological processes. The primary stakeholder for this project is the client, Professor Michelle Kuttel. However, the implications of this work extend beyond a single research group. Biofilms have a significant impact and create challenges in both medical and industrial contexts, making a broad community of researchers, clinicians, and engineers indirect stakeholders. In healthcare, biofilms are associated with persistent infections such as chronic wounds and dental plaque, and they frequently colonize medical implants—leading to severe complications and increased treatment costs. In industry, biofilms contribute to food contamination, reduce equipment lifespan, and create inefficiencies in water treatment systems. As such, this project may be valuable to both academic and applied research communities seeking insight into early-stage biofilm formation.
SMART Goals [5]	Functions and features for users: - Users can adjust several input parameters before running the simulation, to get different results out. These parameters are: - Number and placement of initial bacteria - Bacteria motility force

	- Bacteria reproduction rate - Initial nutrient concentration - EPS production rate - Nutrient threshold for EPS production - EPS size - EPS-cell attraction range - Coefficients of friction between particles - Simulation boundary size - Maximum agent limit - Intermediate output interval: The user can specify for the program to produce an output file at set time intervals. - Users can run the simulation which will calculate the resulting positions of all the bacteria and EPS particles at the end of the simulation and will produce a file which contains the results and can then be visualised. - Users can visualise the results of a finished simulation by running the visualisation model, which uses the output file. Goals: - Simulation runs with no multithreading errors - Simulation can run with a set minimum number of bacteria - Clustering behaviour and size is measured and labelled - Visualisation can be generated from simulation output file - Visualisation is understandable and clear (colour-coordinated correctly) - Interactive forces are correctly implemented
Inputs, outputs, and performance [5]	Inputs: - Number and placement of initial bacteria - Bacteria motility force - Bacteria reproduction rate - Initial nutrient concentration - EPS production rate - Nutrient threshold for EPS production - EPS size - EPS-cell attraction range - Coefficients of friction between particles - Simulation boundary size - Maximum agent limit - Intermediate output interval: The user can specify for the program to produce an output file at set time intervals. Outputs: The user inputs are used to run the simulation, the output of which is written to a separate file. The contents of the output file are then used to visualise the bacterial growth. - Bacteria location - Bacteria orientation - Bacteria size

	- EPS location - EPS size - Depletion effect parameters - o Ecell-cell, Ecell-eps ,Eeps-eps - Time at snapshot (if static) - Particle ID - (Bacteria or EPS) - o Used for colour labelling in visualisation Performance: There are no specific performance requirements for this project as the simulation goal is to be realistic rather than efficient. However, the speed must be reasonable, so that many simulations can be run within a day.
Resources and Constraints [3]	As this is a multithreaded program, the biggest resource requirements are CPU and RAM. A computer with reasonable multithreading capability is required, even then it may end up under strain while running simulations. The model will have limited accuracy due to limited time and complexity constraints. It will not be able to accommodate for all the exact biological factors which exist, as that is an almost impossible level of detail to design a program for, and would be very slow to run. However, the model will aim to reflect enough of the biological factors to provide a reasonably accurate model. Challenging factors also include: All important aspects of the forces acting on every particle must be calculated within this model, and the mathematics must be simple enough that we, the developers, understand the calculations our model does. Also, the model must be designed to efficiently use multithreading, which means that it will be constrained by the need to design with very parallel multithreading in mind.
Feasibility [3]	The scope outlined in this document appears ambitious yet achievable within the allocated timeframe. The core features are well-defined and align with the objectives of simulating early-stage biofilm formation. However, two key concerns remain that may impact feasibility: 1. Scope Creep: Any additional features or requirements introduced after the initial project scoping may exceed the current resource and time constraints. It is important that any changes to the scope be carefully assessed and approved through consultation with the client. 2. Mathematical and Computational Complexity: While we have a general understanding of the modelling approach, certain elements—such as the detailed implementation of EPS dynamics or mechanical interactions—may require deeper mathematical or computational insight than initially anticipated. We will need to assess these complexities early in development. That said, based on our current understanding, we are confident in our ability to deliver a functional and biologically meaningful simulation that meets the core objectives described.